

Features and Modalities I

AI in Music, Winter Term 2025/2026

30.10.2025

Exercise 1 [Extraction and Analysis of Audio Features, 4 points]

For the extraction and analysis of several audio features, please install and configure AMUSE (<https://github.com/AdvancedMUSicExplorer/AMUSE/>).

Two notes for the installation:

- The added JavaFX jars should be of version 17, available at <https://gluonhq.com/products/javafx>.
- For MacOS, the run configuration should be adjusted: in the arguments, “-XstartOnFirstThread” should not be used.

In case you’ll have problems during the configuration of AMUSE, please contact us.

For the three music pieces from the Moodle, please extract the following features with default parameters:

- 0: Zero-crossing rate
- 4: Root mean square
- 6: Low energy
- 14: Spectral centroid
- 32: Angles in phase domain
- 33: Distances in phase domain

Then, write a Python script to visualise the features (plotting windows on the X-axis and feature values on the Y-axis).

Please write a short report (a few sentences), whether the features extracted could be useful for the recognition of musical genres.

Using scipy, estimate Pearson and Spearman correlation coefficients for features zero-crossing rate and spectral centroid.

Exercise 2 [Visualisation of Phase Domain, 2 points]

Please write a Python script to visualise phase domain differences between classical and rock pieces. First, load the signal using librosa package. Then, create a plot with X-Axis showing time signal $x(t)$ and Y-Axis showing time signal $x(t+1)$. Which differences can you see in the phase domains of the two genres?

Exercise 3 [Extraction and Analysis of TF-IDF, 4 points]

The file lyrics50.csv contains lyrics from 50 playlists. Write Python code to estimate TF-IDF for words *always*, *but*, *christmas*, *city*, *feel*, *love*, *think*, *you*, *world*, *yeah*. For the estimation of TF use the following normalisation: the frequency of word occurrence in a song should be divided by the occurrence of this word in all songs. For simplicity, consider lines with “NA” (no lyrics found) as lyrics. Which words have the highest TF-IDF values?

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1b8550bf-b976-4928-a451-711dce932f10;Oh, oh, baby you can walk,
you can talk just like me. You can walk, you can talk just like
me. You can look, tell me what you see. You can look, you won't
see nothing like me. If you look around the world. Oh, baby you
can walk, you can talk just like me. With a little practice you
can walk like, talk just like me. If that's what you wanna do.
Well you can look, you'll walk in circles around me. But first,
I'll walk in circles around you. But first, I'll walk around the
world. I'm walking 'round the world. I. you can. I'll walk in
circles around you. But first, show me what you can do.
00abcd0d-19ca-44dd-b3d5-e44e357b79d9;This and that, they must be
the same. what is legal is just what's real. what I'm given to
understand. is exactly what I steal. I wormed my way into the
heart of the crowd. I wormed my way into the heart of the crowd.
I was shocked to find what was allowed. I didn't lose myself in
the crowd. Shot by both sides. on the run to the outside of
everything. shot by both sides. they must have come to a secret
understanding. New offences always in my nerves. they're taking
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